

# MinImageScorer: Error-Driven Difficulty Scoring for Guided Copy-Paste Augmentation in Agricultural Object Detection

Author Name<sup>1</sup>, Supervisor Name<sup>1</sup>

<sup>1</sup>Department of XXX, University of XXX, Country  
email@university.edu

**Abstract**—Copy-paste augmentation is widely applied to improve object detection but is rarely evaluated against the specific failure mode a dataset exhibits. We propose MinImageScorer, an error-driven framework that scores per-object detection difficulty from baseline model failures using a cost function combining prediction confidence and spatial accuracy, with weights auto-calibrated via Pearson correlation optimization. We validate the score on two agricultural benchmarks — MinneApple and Global Wheat Head (GWHD) — and use it to guide copy-paste augmentation. Experiments across three random seeds show that score-guided selection outperforms random selection on MinneApple ( $\Delta\text{AP} = +0.013$ ) but neither condition recovers the no-augmentation baseline. Feature-space analysis confirms augmented images shift the training distribution away from the test distribution, explaining the consistent degradation. On GWHD, domain shift dominates and copy-paste provides no benefit under either selection strategy. Our findings establish that the utility of copy-paste augmentation depends on the dataset’s underlying failure mode, and that MinImageScorer correctly identifies the objects responsible for that failure.

**Index Terms**—object detection, copy-paste augmentation, difficulty scoring, agricultural detection, small objects, domain shift

## I. INTRODUCTION

Agricultural object detection benchmarks such as MinneApple [1] and Global Wheat Head (GWHD) [2] present distinct structural failure modes. MinneApple is dominated by high-density small objects near the resolution floor of standard backbones, while GWHD contains images from multiple geographic sources with significant cross-domain visual variation. These failure modes differ fundamentally from the object *underrepresentation* problem that Kisantal *et al.* [3] addressed on MS COCO, where copy-paste augmentation produced measurable gains by increasing small object anchor matches during training.

A critical gap remains: practitioners routinely apply copy-paste augmentation without first diagnosing whether the dataset’s failure mode is addressable by adding more object instances. When the failure is structural — resolution-limited detectability or cross-domain shift — copy-paste may corrupt the training distribution rather than improve it. No prior work provides an empirical framework to diagnose this before augmentation is applied.

We make the following contributions:

- 1) **MinImageScorer**: a per-object difficulty score derived from model failure signals, with weights auto-calibrated via Pearson correlation optimization, requiring no domain-specific assumptions.
- 2) **Empirical validation** that the score captures dataset-specific failure modes on both MinneApple and GWHD.
- 3) **Controlled augmentation investigation** comparing score-guided and random copy-paste across three seeds, with feature-space analysis explaining the observed performance patterns.

## II. METHODOLOGY

### A. MinImageScorer: Formulation

Let  $G(I)$  be the ground-truth objects in image  $I$  and  $\mathcal{P}$  the model predictions. For object  $g$ , define  $\mathcal{P}(g; \tau)$  as predictions with  $\text{IoU}(p, g) \geq \tau$ . The *object difficulty score* is:

$$S_{\text{obj}}(g) = \begin{cases} \min_{p \in \mathcal{P}(g; \tau)} [\alpha(1 - \text{conf}(p)) + \beta(1 - \text{IoU}(p, g))], & \mathcal{P}(g; \tau) \neq \emptyset, \\ \alpha + \beta, & \mathcal{P}(g; \tau) = \emptyset. \end{cases} \quad (1)$$

Missed detections receive the maximum score  $\alpha + \beta$ . The *image-level score* aggregates object difficulties and penalises false positives:

$$S_{\text{img}}(I) = w_1 \cdot \frac{1}{|G(I)|} \sum_{g \in G(I)} S_{\text{obj}}(g) + w_2 \cdot \text{FPrate}(I; \gamma, \tau). \quad (2)$$

Weights  $\alpha, \beta, w_1, w_2$  are calibrated by maximizing the Pearson correlation between score distributions and per-image miss rate and false positive rate on the training set.

### B. Score-Guided Copy-Paste Pipeline

The pipeline proceeds as: (1) train a baseline YOLOv8s [4] on the original training set with all built-in augmentations disabled; (2) run inference on the training set to collect confidence and IoU signals; (3) compute  $S_{\text{obj}}$  and  $S_{\text{img}}$  for all objects and images; (4) sample objects proportionally to their scores using exponential weighting ( $\alpha=2$ ) for score-guided, or uniformly for random; (5) paste 2–8 objects per augmented image within the source image (same-image-only); (6) retrain from COCO pretrained weights on the augmented set. Both

TABLE I  
3-SEED MEAN COCO AP ON ORIGINAL TEST SETS. BOLD = BEST  
COPY-PASTE CONDITION PER DATASET.

| Dataset    | Condition       | AP           | AP <sub>50</sub> | AP <sub>75</sub> | AP <sub>S</sub> | AP <sub>M</sub> |
|------------|-----------------|--------------|------------------|------------------|-----------------|-----------------|
| MinneApple | Baseline        | 0.439        | 0.771            | 0.452            | 0.300           | 0.592           |
|            | Random CP       | 0.415        | 0.741            | 0.427            | 0.279           | 0.563           |
|            | <b>Score CP</b> | <b>0.428</b> | <b>0.759</b>     | <b>0.440</b>     | <b>0.286</b>    | <b>0.581</b>    |
| GWHD       | Baseline        | 0.507        | 0.922            | 0.491            | 0.136           | 0.501           |
|            | Random CP       | 0.486        | 0.905            | 0.456            | 0.118           | 0.480           |
|            | <b>Score CP</b> | <b>0.484</b> | <b>0.905</b>     | <b>0.455</b>     | <b>0.098</b>    | <b>0.477</b>    |

conditions are identical in all parameters except the selection policy, ensuring a controlled comparison.

### III. EXPERIMENTS

#### A. Setup

**Datasets.** MinneApple contains images of apple trees with dense, small-object annotations. GWHD contains images of wheat fields collected across multiple geographic sources (Australia, Japan, India, and others).

**Model and training.** All models use YOLOv8s initialized from COCO pretrained weights. Training settings: MinneApple 200 epochs / patience 10; GWHD 100 epochs / patience 8; batch size 16; input size  $1024 \times 1024$ ; learning rate  $10^{-2}$ . All Ultralytics built-in augmentations (mosaic, mixup, HSV jitter, flips) are set to zero to isolate the effect of copy-paste.

**Augmentation.** Dataset ratio 0.3; scale jitter  $[0.9, 1.1]$  (MinneApple),  $[0.8, 1.2]$  (GWHD); no rotation; overlap avoidance with 5 px margin; no blending. Results are reported as means over seeds {123, 456, 789}.

#### B. Results

Table I reports 3-seed mean COCO AP for both datasets.

On MinneApple, score-guided copy-paste outperforms random copy-paste by  $\Delta\text{AP} = +0.013$ , with consistent gains across AP, AP<sub>50</sub>, AP<sub>75</sub>, AP<sub>S</sub>, and AP<sub>M</sub>. This indicates the score successfully directs augmentation toward harder, smaller objects that provide greater gradient signal. However, both copy-paste conditions remain below baseline ( $-0.024$  for Score CP), showing that copy-paste adds distributional noise that outweighs its selective benefit.

On GWHD, score-guided and random selection are statistically indistinguishable (gap:  $-0.002$  AP), and both degrade below baseline ( $-0.023$ ). The failure pattern differs from MinneApple: score-guided sampling concentrates pasted objects from a narrow subset of source domains (predominantly *arvalis*), further amplifying the domain mismatch already present in the dataset rather than addressing it.

#### C. Feature-Space Analysis

Fig. 1 shows image-level backbone embeddings (YOLOv8s P4, global average pool, UMAP projection) for four groups: original training images, augmented images under random CP, augmented images under score CP, and test images. On both datasets, augmented images occupy a region of feature space

FIGURE PLACEHOLDER:  
Image-level UMAP projections for MinneApple (left) and GWHD (right). Groups: original train, augmented (random CP), augmented (score CP), and test images.

Fig. 1. Image-level backbone feature projections (UMAP, P4 layer). Augmented images occupy regions that are systematically farther from the test distribution than original training images, confirming distributional corruption under both augmentation strategies. Score-guided augmented images are displaced further on GWHD due to domain-minority object concentration.

that is displaced from the test distribution relative to the original training images.

Quantitatively, the centroid distance from augmented images to the test distribution exceeds that of original training images on both datasets (MinneApple:  $[value]$  vs  $[value]$ ; GWHD:  $[value]$  vs  $[value]$ ), directly explaining the consistent performance degradation. On GWHD, score-guided augmented images show greater displacement than random augmented images, consistent with domain-minority object concentration.

### IV. DISCUSSION

**Why copy-paste fails on these datasets.** Kisantal *et al.* [3] demonstrated that copy-paste improves AP<sub>S</sub> on MS COCO by increasing anchor matches for underrepresented small objects. This benefit presupposes that: (1) hard objects are above the backbone feature resolution floor, and (2) pasted objects come from the same visual domain as the target scene. Neither condition holds reliably on MinneApple or GWHD. On MinneApple, the hardest objects identified by MinImageScorer are predominantly near the resolution limit of the backbone, making them weakly represented in feature maps regardless of paste frequency. On GWHD, even within-image pasting fails because the feature-space mismatch originates at the image level, not the object level.

**Diagnostic value of the score.** Despite producing no absolute improvement, MinImageScorer demonstrates clear utility: it correctly ranks objects by detectability (score correlates with miss rate;  $r = [value]$  on MinneApple,  $r = [value]$  on GWHD), and on MinneApple it selects objects that produce a  $+0.013$  AP relative gain over uniform random selection. The score identifies the right objects; the limitation is that copy-paste cannot make those objects more learnable when the failure mode is structural.

**Established boundary condition.** Score-guided copy-paste provides a positive relative benefit when the selected objects are above the resolution floor and within-distribution with the target scene. It provides no benefit when the failure mode is domain shift, and it amplifies the degradation on AP<sub>S</sub> when objects

selected are predominantly sub-resolution. This constitutes a practical diagnostic criterion: run MinImageScorer, inspect the size and domain distribution of high-score objects before applying copy-paste augmentation.

## V. CONCLUSION

We proposed MinImageScorer, a model-failure-driven difficulty scoring framework validated on two agricultural detection benchmarks. Controlled experiments show that score-guided copy-paste outperforms random copy-paste on MinneApple (+0.013 AP) but that both strategies degrade below baseline, with the degradation confirmed by feature-space analysis to originate from distributional shift between augmented training images and the test set. On GWHD, domain shift makes copy-paste ineffective under any selection strategy. The practical implication is that copy-paste augmentation should be preceded by failure mode diagnosis: it is effective for underrepresentation (COCO-like datasets) but not for resolution-limited or domain-shifted datasets. Future work includes online augmentation to reduce distributional rigidity and domain-constrained pasting for multi-source benchmarks.

## REFERENCES

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